# CHEMICAL BONDS

## CHEMICAL BONDS

-when two or more atoms combine
-the force that holds the atoms together

### ATOMS FORM CHEMICAL BONDS TO...

-complete their outer energy level.
-become more stable.

OCTET MEANS A GROUP OF EIGHT.

### OCTET RULE:

Simple chemical theory that states that atoms tend to combine in such a way that they each have eight electrons in their valence shells.

## VALENCE ELECTRONS

valence electron
shells

Only electrons in the outer shell (valence electrons) form chemical bonds.

Each shell can only contain a certain number of electrons: the first layer can hold 2, the second and third layer can hold 8

How many valence electrons does each have?

Chlorine Carbon Hydrogen

## THREE MAIN TYPES OF BONDS:

### IONIC

ONE ATOM DONATES ELECTRON(S), THE OTHER ACCEPTS. A BOND IS FORMED BY ATTRACTION BETWEEN NEWLY FORMED IONS. OCCURS BETWEEN METAL AND NON-METAL IONS.

Before After

### COVALENT

Before After

TWO ATOMS SHARE ELECTRONS BETWEEN THEM. OCCURS BETWEEN TWO NON-METALS.

### METALLIC

ATOMS ARE CONNECTED BY A "SEA OF ELECTRONS" OCCURS ONLY BETWEEN METALS.

He
Ne
Ar
Kr
Xe
Rn
Og

Helium

No gas are the most stable elements. They already have full energy levels.

Neon LIGHTS

Argon & Krypton

doodle notes

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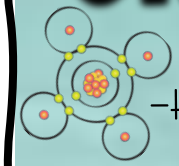
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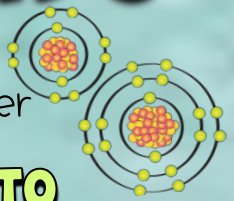
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# CHEMICAL BONDS

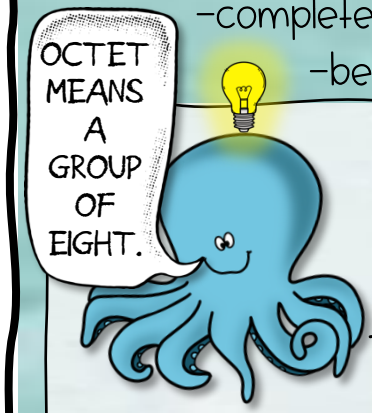


-when two or more atoms combine
-the force that holds the atoms together



## ATOMS FORM CHEMICAL BONDS TO...

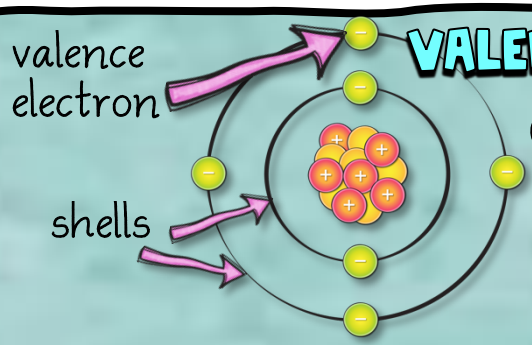
-complete their outer energy level.
-become more stable.




## OCTET RULE:

Simple chemical theory that states that atoms tend to combine in such a way that they each have eight electrons in their valence shells.

## VALENCE ELECTRONS

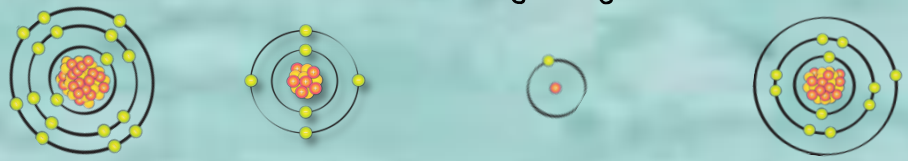



Only electrons in the outer shell (valence electrons) form chemical bonds.

Each shell can only contain a certain number of electrons: the first layer can hold 2, the second and third layer can hold 8.

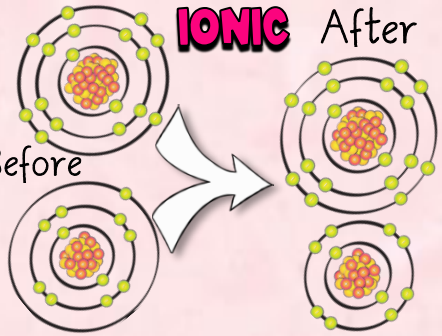
## How many valence electrons does each atom have?

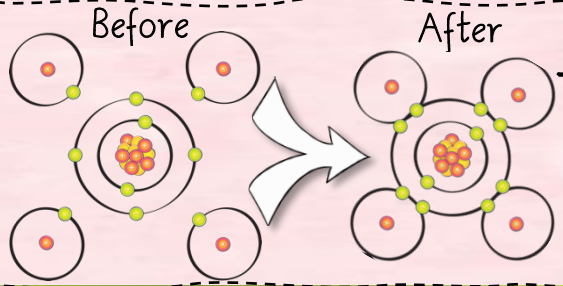
Chlorine ___ Carbon ___ Hydrogen ___ Sodium ___



# THREE MAIN TYPES OF BONDS:

ONE ATOM DONATES ELECTRON(S), THE OTHER ACCEPTS. A BOND IS FORMED BY ATTRACTION BETWEEN NEWLY FORMED IONS. OCCURS BETWEEN METAL AND NON-METAL IONS.



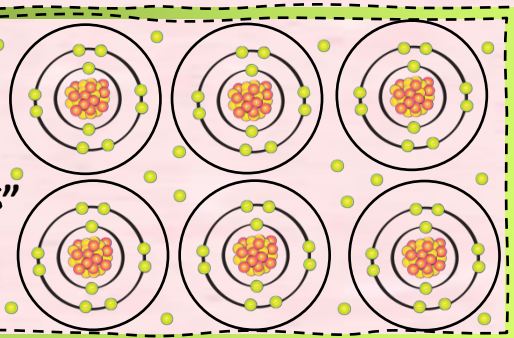




**COVALENT**

TWO ATOMS SHARE ELECTRONS BETWEEN THEM. OCCURS BETWEEN TWO NON-METALS.

**METALLIC**

ATOMS ARE CONNECTED BY A "SEA OF ELECTRONS" OCCURS ONLY BETWEEN METALS.



He
Ne
Ar
Kr
Xe
Rn
Og

Helium



Noble gases are the most stable elements. They already have full energy levels.




Argon & Krypton

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# CHEMICAL BONDS

-when two or more atoms combine

-the force that holds the atoms together

## ATOMS FORM CHEMICAL BONDS TO...

-complete their outer energy level.

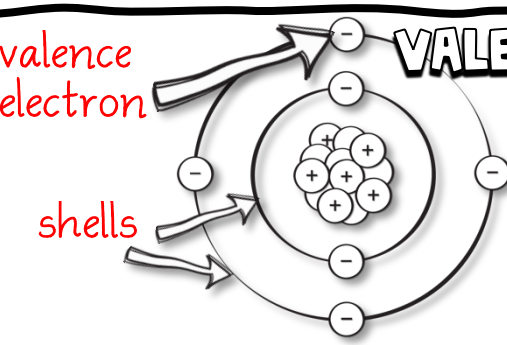
-become more stable.

OCTET MEANS A GROUP OF EIGHT.

## OCTET RULE:

Simple chemical theory that states that atoms tend to combine in such a way that they each have eight electrons in their valence shells.

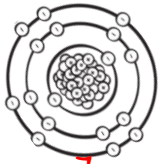
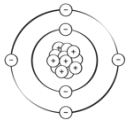
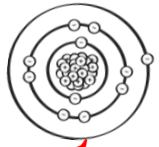
## VALENCE ELECTRONS




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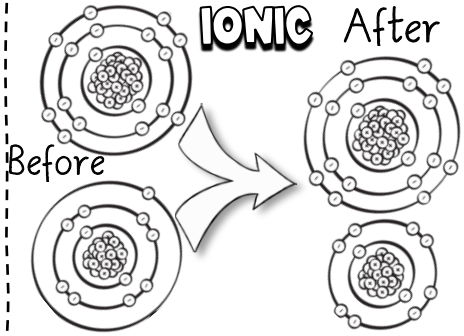
### How many valence electrons does each atom have?

| Chlorine | Carbon | Hydrogen | Sodium |
|---|---|---|---|
|  |  |  |  |
| 7 | 4 | 1 | 1 |

## THREE MAIN TYPES OF BONDS:

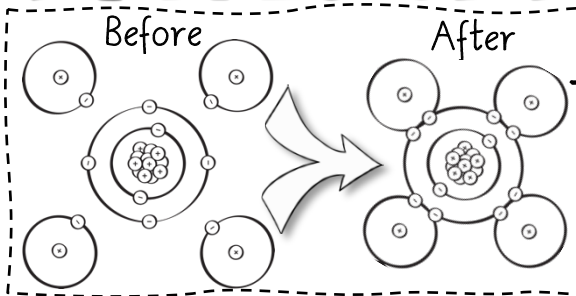
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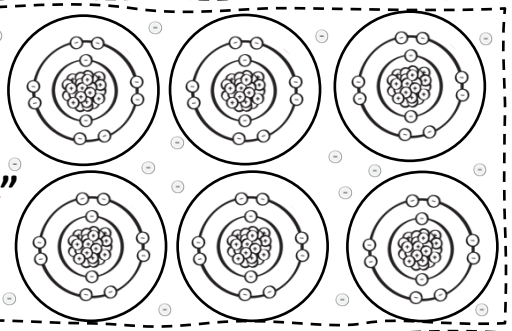

### COVALENT

TWO ATOMS SHARE ELECTRONS BETWEEN THEM. OCCURS BETWEEN TWO NON-METALS.




### METALLIC

ATOMS ARE CONNECTED BY A "SEA OF ELECTRONS" OCCURS ONLY BETWEEN METALS.



| |
|---|
| He |
| Ne |
| Ar |
| Kr |
| Xe |
| Rn |
| Og |

Helium

Noble gases are the most stable elements. They already have full energy levels.






Argon & Krypton

# CHEMICAL BONDS

-when _____ atoms combine

-the _____ that holds the atoms together

## ATOMS FORM CHEMICAL BONDS TO...

-complete their _____ energy level.

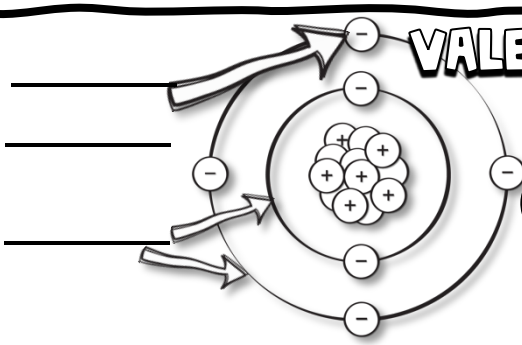
-become more _____.

OCTET MEANS A GROUP OF EIGHT.

## OCTET RULE:

Simple chemical _____ that states that atoms tend to _____ in such a way that they each have ____ electrons in their _____ shells.

## VALENCE ELECTRONS



_____

_____

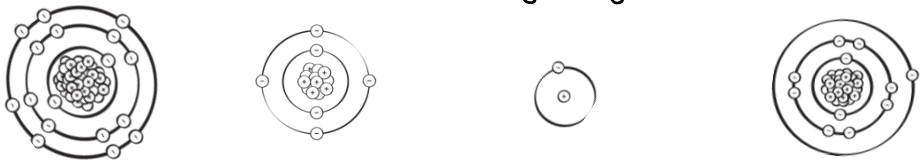
_____

Only electrons in the outer shell (_____ _____) form chemical bonds.

Each shell can only contain a _____ of electrons: the first layer can hold ___, the second and third layer can hold ___.

### How many valence electrons does each atom have?

Chlorine Carbon Hydrogen Sodium

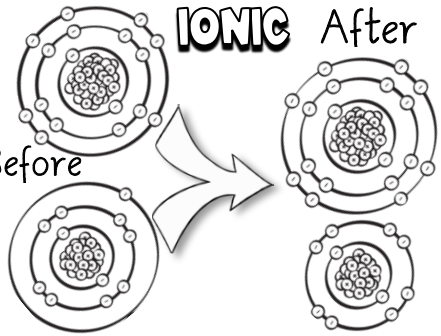


___ ___ ___ ___

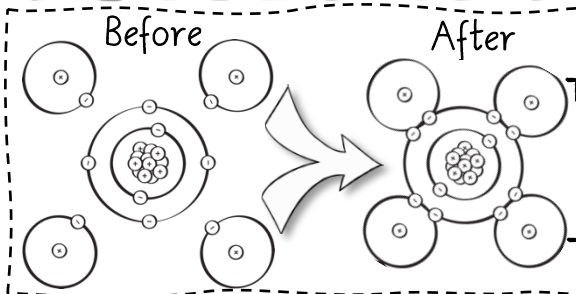
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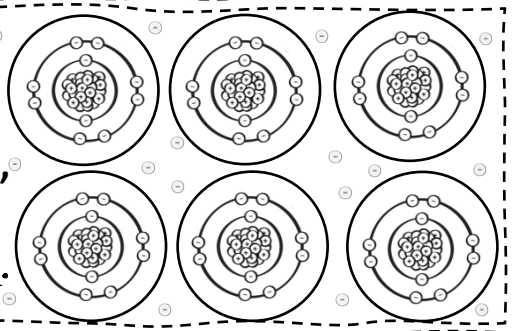

### COVALENT




TWO ATOMS _____ ELECTRONS BETWEEN THEM. OCCURS BETWEEN TWO _____.

### METALLIC

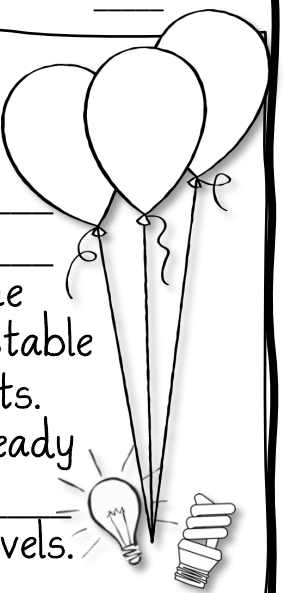
ATOMS ARE CONNECTED BY A "_____" OCCURS ONLY BETWEEN _____.



| |
|---|
| He |
| Ne |
| Ar |
| Kr |
| Xe |
| Rn |
| Og |

Helium

_____ _____ are the _____ stable elements. They already have _____ energy levels.



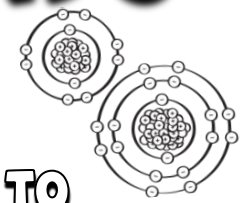
Neon LIGHTS

Argon & Krypton

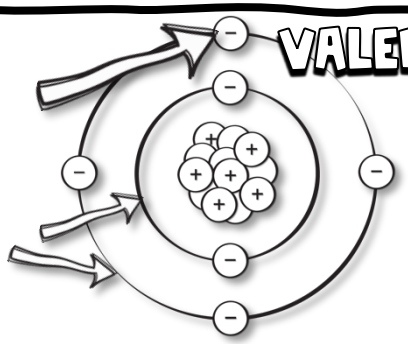
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# CHEMICAL BONDS





## VALENCE ELECTRONS



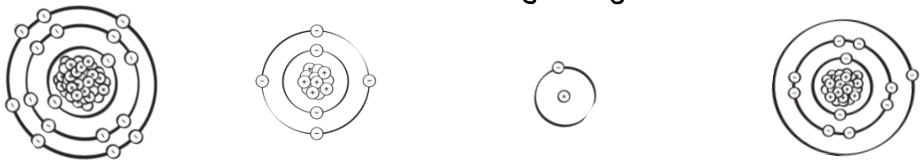
## ATOMS FORM CHEMICAL BONDS TO...




## OCTET RULE:

## How many valence electrons does each atom have?

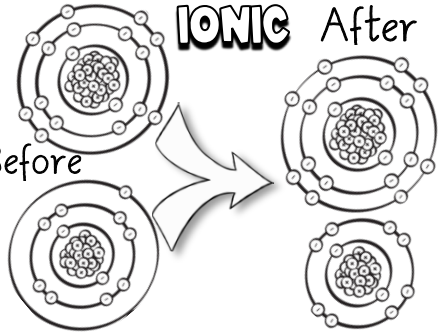
Chlorine ___ Carbon ___ Hydrogen ___ Sodium ___

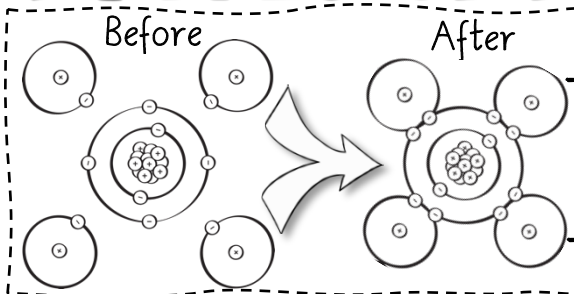


## THREE MAIN TYPES OF BONDS:

ONE ATOM _____ ELECTRON(S), THE OTHER _____. A BOND IS FORMED BY _____ BETWEEN NEWLY FORMED _____. OCCURS BETWEEN _____ AND _____ IONS.

### IONIC



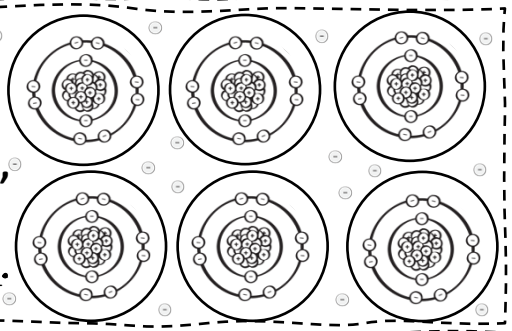




### COVALENT

TWO ATOMS _____ ELECTRONS BETWEEN THEM. OCCURS BETWEEN TWO _____.

### METALLIC

ATOMS ARE CONNECTED BY A "_____" OCCURS ONLY BETWEEN _____.



| He |
| --- |
| Ne |
| Ar |
| Kr |
| Xe |
| Rn |
| Og |

Helium



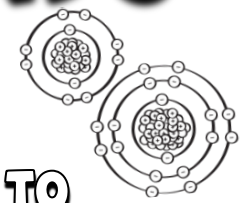



Argon & Krypton

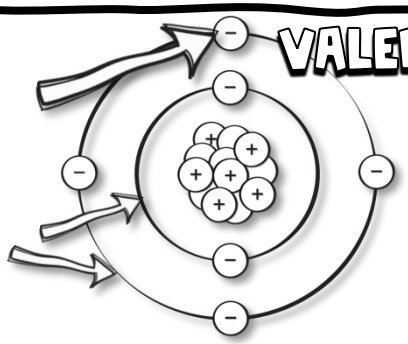
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# CHEMICAL BONDS

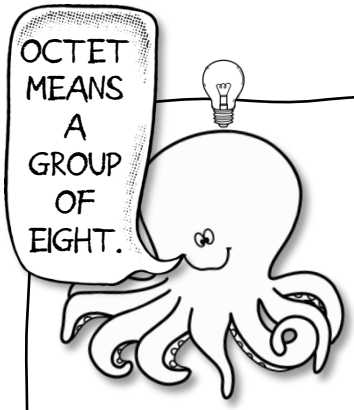




## VALENCE ELECTRONS

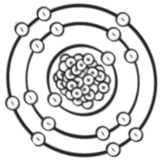
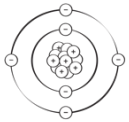
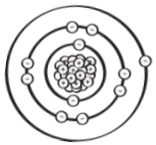


## ATOMS FORM CHEMICAL BONDS TO...



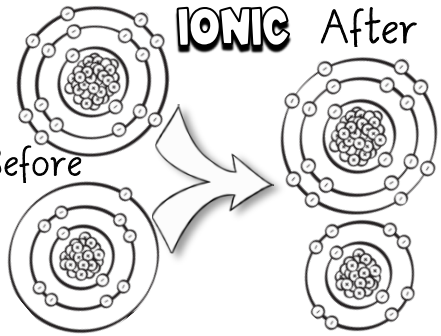

## OCTET RULE:

How many valence electrons does each atom have?

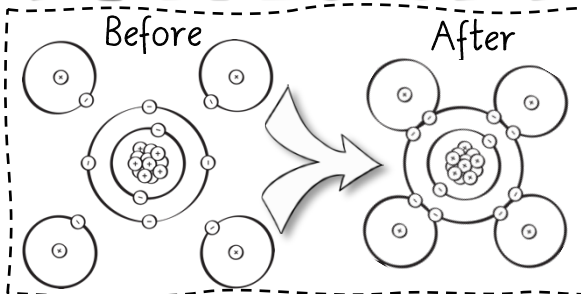
| Chlorine | Carbon | Hydrogen | Sodium |
|---|---|---|---|
|  |  |  |  |
| ___ | ___ | ___ | ___ |

## THREE MAIN TYPES OF BONDS:

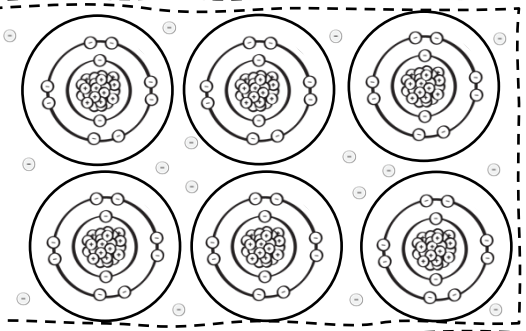
### IONIC




### COVALENT

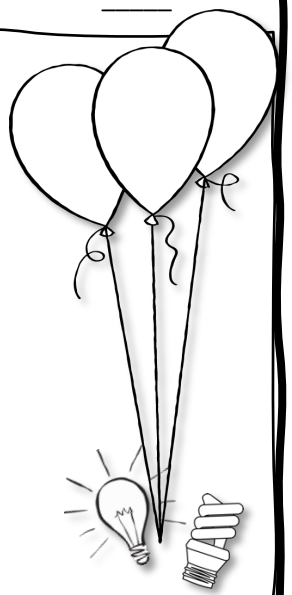



### METALLIC



| |
|---|
| He |
| Ne |
| Ar |
| Kr |
| Xe |
| Rn |
| Og |

Helium

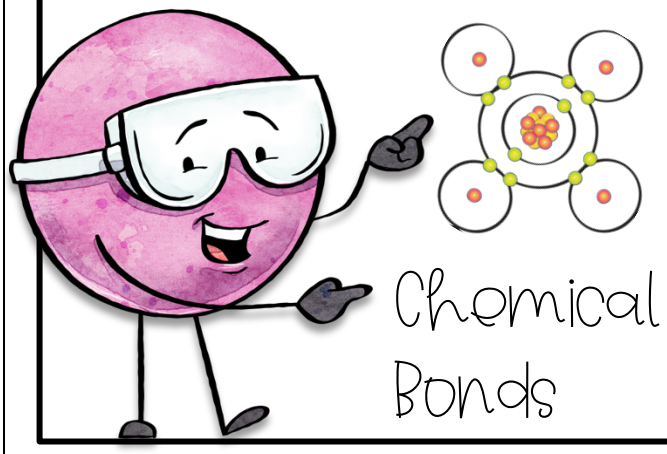





Argon & Krypton

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# Understanding Checkpoint



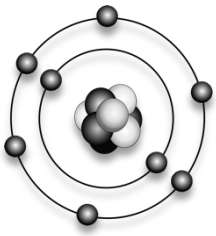

Name _____

Date _____

Period _____

1. Chemical bonds occur when _____ or more atoms combine.

2. How many valence electrons are represented here?



_____

3. Which is involved in bonding?
   a. protons b. neutrons
   c. electrons d. valence electrons

4. In an ionic bond, electrons are _____.
   a. lost or gained
   b. shared
   c. shared or gained
   d. none correct

5. In a covalent bond, electrons are _____.
   a. lost or gained
   b. shared
   c. shared or gained
   d. none correct

6. Which of the following has a full valence shell?
   a. Oxygen b. Helium
   c. Barium d. Carbon

7. Typically, atoms are more stable when they are
   a. bonded together.
   b. apart from each other.
   c. metal ions.
   d. all of these

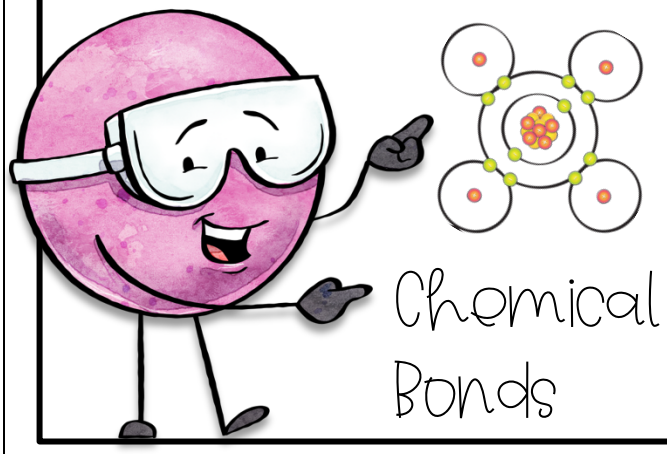
8. What two types of atoms make a covalent bond?
   a. 2 Nonmetals
   b. 1 Nonmetal and 1 Metal
   c. 2 Metals
   d. 2 Noble Gases

9. How many valence electrons does Neon have?
   a. 2 b. 3
   c. 8 d. 18

10. A metallic bond is a bond between
    a. valence electrons and positively charged metal ions.
    b. the ions of two different metals.
    c. a metal and nonmetal.
    d. none of the above

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# Understanding Checkpoint



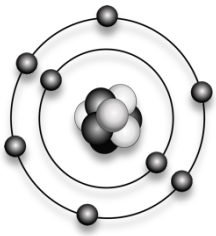

Name _____

Date _____ KEY

Period _____

1. Chemical bonds occur when __2__ or more atoms combine.

2. How many valence electrons are represented here?



__6__

3. Which is involved in bonding?
   a. protons  b. neutrons
   c. electrons  d. valence electrons

4. In an ionic bond, electrons are _____.
   a. lost or gained
   b. shared
   c. shared or gained
   d. none correct

5. In a covalent bond, electrons are _____.
   a. lost or gained
   b. shared
   c. shared or gained
   d. none correct

6. Which of the following has a full valence shell?
   a. Oxygen  b. Helium
   c. Barium  d. Carbon

7. Typically, atoms are more stable when they are
   a. bonded together.
   b. apart from each other.
   c. metal ions.
   d. all of these

8. What two types of atoms make a covalent bond?
   a. 2 Nonmetals
   b. 1 Nonmetal and 1 Metal
   c. 2 Metals
   d. 2 Noble Gases

9. How many valence electrons does Neon have?
   a. 2  b. 3
   c. 8  d. 18

10. A metallic bond is a bond between
    a. valence electrons and positively charged metal ions.
    b. the ions of two different metals.
    c. a metal and nonmetal.
    d. none of the above

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